Chapter 12 • Information Handling

Solved Review Exercise 12

MCQs

1. Which data takes only specific values?
a) continuous data
b) discrete data
c) grouped data
d) ungrouped data
Ans: b
Discrete data take countable, separated values.

2. The number of times a value occurs is its:
a) frequency
b) relative frequency
c) class limit
d) class boundary
Ans: a
This is the definition of frequency.

3. Midpoint is also known as:
a) mean
b) median
c) class limit
d) class mark
Ans: d
Class mark is the midpoint of a class.

4. A frequency polygon can also be constructed using a:
a) histogram
b) bar graph
c) class boundary
d) class limit
Ans: a
Join histogram bar-top midpoints and extend to zero.

5. Greatest value minus smallest value is the:
a) class limits
b) midpoint
c) relative frequency
d) range
Ans: d
This is the range.

6. A measure of central tendency finds the data set's:
a) class boundaries
b) cumulative frequency
c) middle or centre value
d) frequency
Ans: c
Mean, median and mode locate a centre.

7. If the mean of 5,7,8,9,x is 7.5, then x=
a) 10
b) 8
c) 8.5
d) 5.8
Ans: c
Sum required=37.5; known sum=29, so x=8.5.

8. Mode of 2,5,8,9,0,1,3,7,10 is:
a) 5
b) 7
c) 0
d) no mode
Ans: d
Every value occurs once.

9. The observation occurring most often is the:
a) mean
b) mode
c) median

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d) weighted mean
Ans: b
This defines the mode.

10. Median of 110,125,122,130,124,127,120 is:
a) 124
b) 120
c) 125
d) 127
Ans: a
Sorted middle value: 110,120,122,124,125,127,130.

2. Define frequency distribution, unequal-class histogram, mean and median.
(i) A frequency distribution lists values/classes with the number of observations in each.
(ii) An unequal-class histogram uses touching bars with class widths on the x-axis and frequency density on the y-axis, so bar area=frequency.
(iii) Mean=sum of observations/number of observations; grouped mean=Σfx/Σf.
(iv) Median is the middle ordered observation, or the mean of the two middle observations for even n.

3. The 40 weights (lb) are: 138,164,150,132,144,125,149,157,146,158,140,147,136,148,152,144; 168,126,138,176,163,119,154,165,146,173,142,147,135,153,140,135; 161,145,135,142,150,156,145,128. (a) Make six classes, (b) draw a histogram and (c) draw a frequency polygon.
Minimum=119,maximum=176; six classes of width 10 give 119–128 through 169–178.
Counting the raw observations gives frequencies 4,7,13,9,5,2; total=40.
Histogram boundaries are 118.5–128.5, ..., 168.5–178.5.
Frequency-polygon midpoints=123.5,133.5,143.5,153.5,163.5,173.5; extend to zero at 113.5 and 183.5.

4. Draw a frequency polygon on the histogram for weights 50–56,57–59,60–64,65–72,73–75,76–80 with frequencies 25,32,40,30,15,8.
Boundaries=49.5–56.5,56.5–59.5,59.5–64.5,64.5–72.5,72.5–75.5,75.5–80.5.
Widths=7,3,5,8,3,5; densities≈3.571,10.667,8,3.75,5,1.6. Join the histogram midpoint densities as in Figure 2.
Unequal-class density histogram and polygon shown in Figure 2.

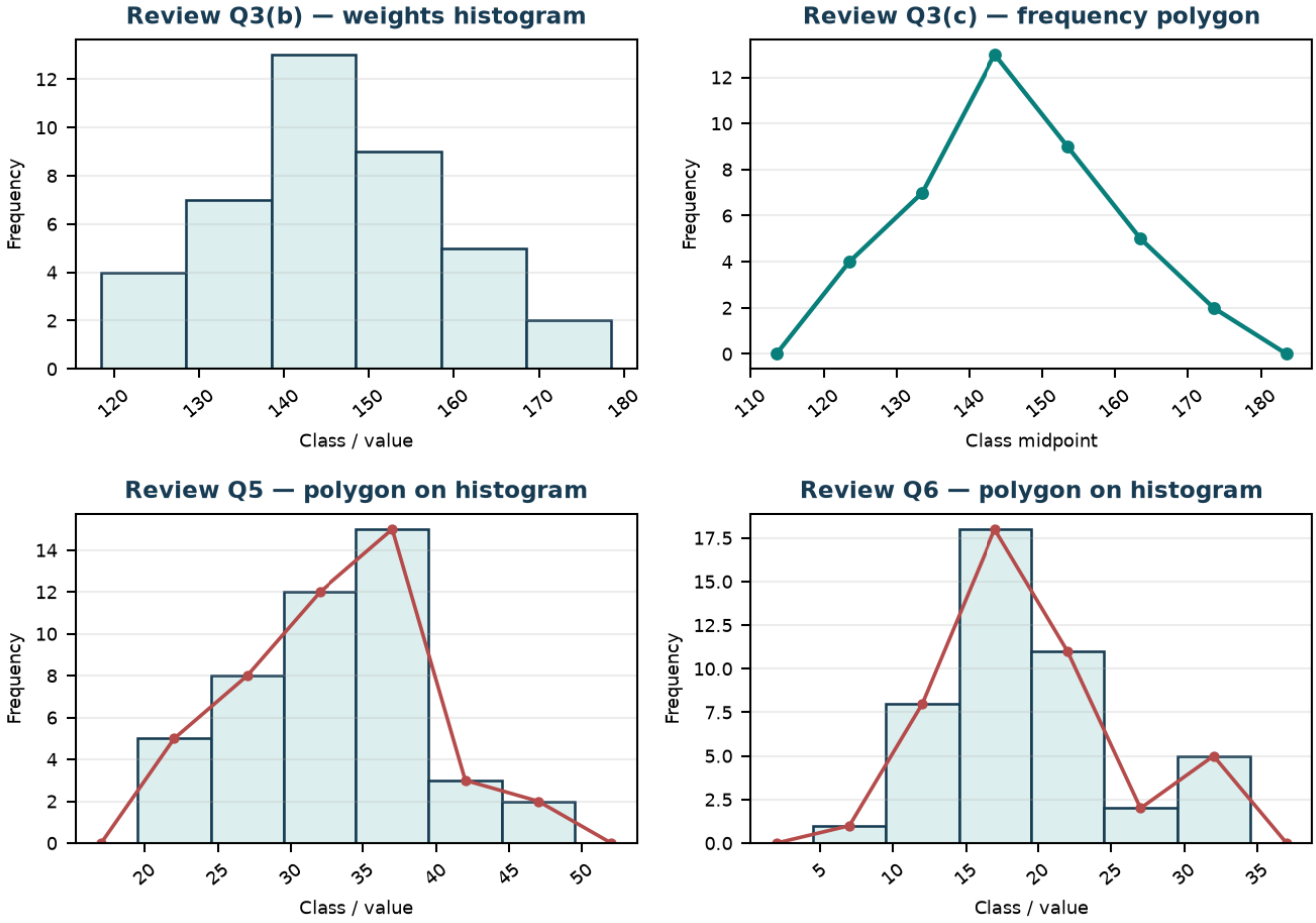
5. For Biology marks classes 20–24 through 45–49 with frequencies 5,8,12,15,3,2, answer the class questions and graph the data.
(i) Fifth class = 40–44, so upper class boundary = 44.5.
(ii) Lower class boundaries=19.5,24.5,29.5,34.5,39.5,44.5.
(iii) Midpoints=22,27,32,37,42,47.
(iv) Least-frequency class=45–49 with frequency 2.
Draw adjacent histogram bars and join midpoints with a frequency polygon, extending to zero at 17 and 52.

Marks	Boundaries	Midpoint	Frequency
20–24	19.5–24.5	22	5
25–29	24.5–29.5	27	8
30–34	29.5–34.5	32	12
35–39	34.5–39.5	37	15
40–44	39.5–44.5	42	3
45–49	44.5–49.5	47	2

6. Draw a frequency polygon and histogram for classes 5–9 through 30–34 with frequencies 1,8,18,11,2,5.

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Use boundaries 4.5–9.5, ..., 29.5–34.5 and midpoints 7,12,17,22,27,32. Join bar-top midpoints and extend to zero at 2 and 37; total frequency=45.



7. Find weighted mean for 20 chairs@500, 20 tables@400, 10 blackboards@750, 25 tube lights@230, 9 cupboards@950.

Total quantity=84; weighted sum=10,000+8,000+7,500+5,750+8,550=39,800.

Weighted mean=39,$\frac{800}{84}$=Rs. 473.81.

8. Rs. 50,000 is allocated as 15,000,12,000,6,000,10,000,7,000 across five sectors.

Average = 50,$\frac{000}{5}$

= Rs. 10,000 per sector.

9. Saad's six test marks are 84,91,72,68,87,78.

Mean=$\frac{84+91+72+68+87+78}{6}=\frac{480}{6}$=80 marks.

10. Maximum-load classes 93–97 through 118–122 have frequencies 2,5,8,12,6,2. Find mean by short method.

Midpoints=95,100,105,110,115,120

choose A = 110. Deviations D

= −15,−10,−5,0,5,10.

Σf=35

ΣfD = −30−50−40+0+30+20

= −70. $\bar{x}$

= 110−$\frac{70}{35}$

= 108.

Mean load=108 kg.

11. Usman recorded dice sums 8,5,6,6,9,4,3,11. Find median and mode.

Sorted 3,4,5,6,6,8,9,11: median = $\frac{6+6}{2}$

= 6; mode
= 6.

12. Wage classes 600–700,700–800,800–900,900–1000,1000–1100 have employee frequencies 3,5,7,21,11. Find mean, median and mode.

Midpoints=650,750,850,950,1050; Σf=47 and Σfx=43,150. Mean=43,$\frac{150}{47}$=918.085.
Median class = 900–1000: L
= 900,h
= 100,cf
= 15,f
= 21,$\frac{n}{2}$
= 23.5. Median
= 900+[$\frac{23.5-15}{21}$]100
= 940.476.
Mode = 900+[$\frac{21-7}{42-7-11}$]100
= 958.333.
Mean≈Rs. 918.09; median≈Rs. 940.48; mode≈Rs. 958.33.

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